

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I SK. Arshad Basha (192212336) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

SK. Arshad Basha

23+24+22+18+8
= 93

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks (100)
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly	23
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	24
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	20
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	18
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	8

99 100

(a) Propositional Logic Representation

F = Patient has Fever

C = Patient has Cough

B = Patient has Breathless ness

m = Patient has measles

P = Patient has pneumonia

For Patient A:-

FA = Patient A has fever

CA = Patient A has cough

BA = Patient A has Breathless ness.

Given

FA = TRUE

CA = TRUE

BA = TRUE.

(b)

Forward Chaining starts with known facts and repeatedly applied rules

FA, CA, BA

$F \rightarrow F \vee C$

FA = TRUE, CA = TRUE.

$\therefore F \vee C = \text{TRUE}.$

$$CB = P$$

$$CA = \text{True}, BA = \text{True}$$

$$PA = \text{True}$$

$$FA \times \text{RashA}$$

But:

RashA is not given

Final forward Gaining facts

FA, CA, BA, FluA, PneumoniaA

(c)

PneumoniaA

Find a rule whose conclusion is Pneumonia

$$CB \rightarrow P$$

$$CA \times BA$$

$$\therefore CA = \text{True}$$

$$BA = \text{True}$$

Hence:

$$\text{PneumoniaA} = \text{True}$$

Question-2

Autonomous Traffic management Agent

Given knowledge Base

Vehicle (x). Emergency Type (x) $\rightarrow$ clear path

clear path (x). Green signal (x) $\rightarrow$ Red signal

Vehicle(x) $\times$ Emergency Type(x) $\rightarrow$ Clear path

Facts

Vehicles(Ambulance)

Emergency Type(Ambulance)

Compare:-

Vehicle(x) = vehicle(Ambulance)

Emergency Type(x) = Emergency Type(Ambulance)

$\Theta = \{x / \text{Ambulance}\}$

(b) Vehicle(Ambulance) $\times$ Emergency Type
Clear Path(Ambulance)

Clear Path(x) $\rightarrow$ Green signal(x) $\rightarrow$ Red signal

Clear Path(Ambulance) $\rightarrow$ Green signal

Vehicle(x) $\times$ Behind(x,y) $\times$ Green signal(y)

Vehicle(x) $\vee \rightarrow$ Behind(x,y)

x = Car A, y = Ambulance

Therefore:-

Proceed(Car A) is TRUE

(c)

The current KB is partially sufficient but not sufficient

for complete simultaneous

① Emergency priority rules

② Collision avoidance

③ Traffic-light coordination.

For Example:-

Vehicle (Fire truck)

Emergency type (Fire truck)

Behind (Fire truck, ambulance)

Emergency type (x) x priority (x,y) $\rightarrow$ proceed.

③ Agricultural Expert Reasoning system

① CNF conversion

P1

Soil dry $\rightarrow$ Irrigation needed

Soil dry x Irrigation needed

P2

Irrigation need x crop wheat $\rightarrow$ Apply drip

CNF:-

Irrigation needed $\vee$ crop wheat

Facts

- ① soil dry
- ② crop wheat

⑥ Resolution proof

Goal:-

Apply drip method

Negated goal:-

Apply drip method

From

soil dry x Irrigation needed

we derive

soil dry

Irrigation needed

Irrigation needed x crop wheat x Apply drip
crop wheat.

∴ KB $\Rightarrow$ Apply drip method.

⑦ Remove soil dry

After removing

soil dry

we still have

crop wheat

But rule #1

soil dry $\rightarrow$ Irrigation needed

$\therefore$ Irrigation needed cannot be derived

crop At Risk is also not provable.

④ Question 4 $\rightarrow$ wumpus world

Given percepts

stench [1,2]

Breeze [1,1]

Glitter [2,2]

⑥ Belief state

From stench [1,2]

[1,1], [1,3], [2,2]

$\therefore \text{wumpus} \in \{[1,1], [1,3], [2,2]\}$

From Breeze [1,1]

$$= \{(1,2), (2,1)\}$$

⑥ Forward Chaining

Stench $[1,2]$



wumpus adjacent



$[1,2], [1,3], [2,2]$



Breeze $[1,1]$



Pit adjacent



$[1,3], [2,1]$



Gold $[2,2]$

Hence

$$\text{wumpus} = \{(1,1), (1,3), (2,2)\}$$

$$\text{pit} = \{(1,2), (2,1)\}$$

$$\text{Gold} = (2,2)$$

⑦ IS $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ safe?

for $[1,2]$

Breeze $[1,1]$

we know a pit may be adjacent.

Pit $[1,2]$

There $[1,2]$ is potentially dangerous

$[2,2]$

stench $[1,2]$

wumpus may be $[2,2]$

$\therefore$ The path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is not a proven

The agent should obtain additional percepts before moving through the uncertain cells.